

MAHMOUD ABDELNABY

SOC Analyst

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PROFESSIONAL SUMMARY

Computer & Systems Engineering graduate with 3+ years of hands-on information security practice, targeting SOC Analyst and DFIR positions. Conducted structured digital forensics investigations including Windows event log/Sysmon analysis, PCAP-based network forensics, MITRE ATT&CK mapping, and timeline reconstruction across simulated APT scenarios. Proficient in Linux/Windows administration, security monitoring automation with Bash/PowerShell, and AWS cloud infrastructure. Supported by practical penetration testing, malware analysis foundations, and full-stack engineering experience.

TECHNICAL SKILLS

Security Operations & Digital Forensics: SOC Tier 1/2 · Alert Triage · Incident Investigation · SIEM (Splunk, ELK, Wazuh) · IOC Analysis · MITRE ATT&CK · Windows Forensics (Event Logs, Sysmon, Registry) · Network Forensics (Wireshark, tshark) · Email Forensics · Timeline Reconstruction · DFIR Reporting

Penetration Testing: Network Enumeration · Web Security Testing · Nmap · Burp Suite · Metasploit · Hydra · Hashcat · SQLmap

Malware Analysis & Reverse Engineering: Static Analysis · PE Analysis · x86/x64 Assembly · Ghidra · WinDbg · GDB

Systems & Networking: Linux (RHEL, Ubuntu, Kali) · ssh · TCP/IP · DNS · DHCP · VLANs · Routing & Switching · NAT · ACLs · systemd · journalctl

Cloud & DevOps: AWS (EC2, S3, Lambda, VPC, IAM, API Gateway, CloudFront) · Docker · Git · Bash · PowerShell · Terraform · Ansible

Programming: Python · Bash · PowerShell · C++ · Java · JavaScript · TypeScript · Assembly (x86/x64)

PROJECTS & TECHNICAL EXPERIENCE

Digital Forensics & Incident Response Investigations

Hands-on Security Labs — [Reports](#) | [Mahmoud Abdelnaby](#)

Conducted end-to-end DFIR investigations on simulated incidents, reconstructing full attack chains and documenting findings using structured incident response methodology.

- Investigated an APT spear-phishing intrusion — analyzed email headers, PCAP captures, and PowerShell Script Block logs (EventID 4104) to trace a PowerShell C2, credential theft via Sticky Notes/KeePass, and DNS data exfiltration
- Reconstructed a CVE-2022-30190 (Follina) compromise chain through Sysmon telemetry (Event IDs 1, 3, 11, 22) and Windows Event Logs — covering multi-stage payload delivery, Chisel SOCKS proxy, PrintSpoofer privilege escalation to SYSTEM, and Windows service persistence
- Mapped adversary behavior to MITRE ATT&CK (20+ techniques) across both investigations, extracted IOCs (domains, IPs, hashes), and built minute-level timelines using Timeline Explorer and SysmonView
- Authored a DFIR methodology reference covering evidence acquisition, volatile/non-volatile analysis, and forensic reporting per NIST SP 800-86 and SP 800-61

Tools: Wireshark · tshark · Sysmon · SysmonView · Timeline Explorer · Inkpase · VirusTotal

Infrastructure & Security Monitoring Automation

Systems Automation & Cloud Engineering

- Developed a Bash-based Linux security monitoring script automating failed SSH authentication detection, sudo privilege escalation auditing, sudoers enumeration, service status checks (sshd, docker, ufw), system log analysis via journalctl, and network health verification
- Automated AWS infrastructure provisioning — VPC with public/private subnets, NACLs, route tables, and internet gateway via idempotent Bash/AWS CLI scripts
- Built a serverless deployment script provisioning DynamoDB, Lambda, API Gateway, S3, IAM (least-privilege policies), and ACM TLS certificates

Tools: Bash · AWS CLI · journalctl · systemd · ss · mpstat · sar

Qanoony — Legal Services Platform

Graduation Project — Zagazig University

Legal services platform connecting clients with lawyers. Contributed backend functionality for the Search, Services, Payments, and Consultations modules.

- Payments: Integrated Stripe Checkout with webhook signature verification; enforced ownership authorization, duplicate payment prevention, and automated refund handling
- Consultations: Developed backend logic for booking management, implementing role-based access control (Client/Lawyer/Admin) and state transition validation
- Services: Built service catalog APIs with comprehensive Zod schema input validation for data integrity and payment-linked request confirmation
- Search: Implemented a cross-entity search API with regex-escaped input sanitization and role-scoped result filtering for secure data retrieval

Stack: Node.js · TypeScript · Express.js · MongoDB · Stripe · Zod

SnapLoad — Desktop Video Downloader

[Mabdelnaby1924/SnapLoad: A modern desktop GUI for yt-dlp](#)

Identified a personal use case and designed a desktop GUI application wrapping yt-dlp. Defined the requirements, workflow, and architecture; used AI-assisted development for implementation.

- Designed the application architecture with separated core logic and UI layers, threaded download handling, real-time progress tracking, and error parsing
- Built and packaged as a standalone Windows executable with Python/CustomTkinter

Stack: Python · CustomTkinter · yt-dlp · PyInstaller

EDUCATION

Bachelor of Science (B.Sc.) in Computer & Systems Engineering

Zagazig University — Faculty of Engineering | 2026

CERTIFICATIONS & COURSES

CCNA

RHCSA — Self-Study (*Red Hat System Administration*)

SOC Level 1 & Level 2 — Learning Path Completed (*TryHackMe*)

Jr Penetration Tester — Learning Path Completed (*TryHackMe*)

OSCP — Self-Study

AWS Cloud Practitioner — Self-Study (*AWS Foundational*)